

HumanEval TRAIN-set typicality metrics — per-epoch dynamics

gemma-4-31b × s2 (RankAlign) & s4 (self-tc) × base/ep0/ep1/ep2

2026-06-19

One table per metric (gen ROC, val ROC, val acc, Pearson). Each shows the **raw** and **tc** (typicality-corrected) gen variants, with an **epoch axis** (base → ep0 → ep1 → ep2) so the training dynamics are visible. Systems: gemma-4-31b, settings s2 (RankAlign) and s4 (self-tc / New+fsx+self-tc), on both `humaneval-v2.1correct-upper` and `-multi`. Values ×100, 1 decimal; means over the 82 problems. **Scores are on the --train split** (the candidate pool the models were trained against), subsampled to N=50 *stratified* candidates per problem (25 positive / 25 negative). Eval = (typicality mode)/(base-typ vs no-base).

Notes: (i) **val ROC** and **val acc** are validator-based, so their raw and tc columns are identical by construction (shown for uniformity). (ii) **raw** gen scores do not depend on the eval mode, so the raw column repeats within a (dataset, setting, epoch) block. (iii) The **base** row is the untrained gemma-4-31b-it model (shared start point for both settings). (iv) s4 (self-tc) was evaluated in both *self* (**self/no-base**) and *base*-typicality (**self/base-typ**) — i.e. self + base — but not in the neg-typicality modes; base and s2 were evaluated in all four modes (the neg-mode absence for s4 is by design, not a gap). (v) These are TRAIN-split numbers — generator/validator agreement is expected to be high on data the model was trained on; the interest is the *dynamics* (how raw vs tc move across epochs), not the absolute level. The companion TEST-split tables are in `docs/he_harvest.2026-06-17/`.

Table 1: HumanEval TRAIN-set gen ROC (×100) per-epoch dynamics, raw vs tc gen variant. gemma-4-31b, s2 (RankAlign) & s4 (self-tc), base/ep0/ep1/ep2, both datasets. Means over 82 problems, N=50 stratified candidates each.

Data	Setting	Epoch	Eval	raw	tc
upper	base	base	self/base-typ	74.1	68.8
upper	base	base	self/no-base	74.1	68.8
upper	base	base	neg/base-typ	74.1	72.2
upper	base	base	neg/no-base	74.1	72.2
upper	s2	ep0	self/base-typ	96.5	81.6
upper	s2	ep0	self/no-base	96.5	85.3
upper	s2	ep0	neg/base-typ	96.5	85.9
upper	s2	ep0	neg/no-base	96.5	87.5
upper	s2	ep1	self/base-typ	94.4	80.0
upper	s2	ep1	self/no-base	94.4	82.7
upper	s2	ep1	neg/base-typ	94.4	85.4
upper	s2	ep1	neg/no-base	94.4	80.8
upper	s2	ep2	self/base-typ	96.3	79.4
upper	s2	ep2	self/no-base	96.3	83.4
upper	s2	ep2	neg/base-typ	96.3	83.4
upper	s2	ep2	neg/no-base	96.3	87.2
upper	s4	ep0	self/base-typ	93.1	87.4
upper	s4	ep0	self/no-base	93.1	83.5
upper	s4	ep1	self/base-typ	91.7	90.9
upper	s4	ep1	self/no-base	91.7	84.5
upper	s4	ep2	self/base-typ	91.0	90.9
upper	s4	ep2	self/no-base	91.0	86.1
multi	base	base	self/base-typ	53.0	64.6
multi	base	base	self/no-base	53.0	64.6
multi	base	base	neg/base-typ	53.0	66.9
multi	base	base	neg/no-base	53.0	66.9

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Data	Setting	Epoch	Eval	raw	tc
multi	s2	ep0	self/base-typ	98.9	95.4
multi	s2	ep0	self/no-base	98.9	95.8
multi	s2	ep0	neg/base-typ	98.9	95.7
multi	s2	ep0	neg/no-base	98.9	84.3
multi	s2	ep1	self/base-typ	98.1	92.0
multi	s2	ep1	self/no-base	98.1	94.1
multi	s2	ep1	neg/base-typ	98.1	94.7
multi	s2	ep1	neg/no-base	98.1	88.5
multi	s2	ep2	self/base-typ	94.6	95.2
multi	s2	ep2	self/no-base	94.6	90.4
multi	s2	ep2	neg/base-typ	94.6	94.6
multi	s2	ep2	neg/no-base	94.6	78.7
multi	s4	ep0	self/base-typ	85.1	97.9
multi	s4	ep0	self/no-base	85.1	89.1
multi	s4	ep1	self/base-typ	86.4	98.2
multi	s4	ep1	self/no-base	86.4	89.9
multi	s4	ep2	self/base-typ	85.6	99.0
multi	s4	ep2	self/no-base	85.6	94.2

Table 2: HumanEval TRAIN-set val ROC ($\times 100$) per-epoch dynamics, raw vs tc gen variant. gemma-4-31b, s2 (RankAlign) & s4 (self-tc), base/ep0/ep1/ep2, both datasets. Means over 82 problems, N=50 stratified candidates each.

Data	Setting	Epoch	Eval	raw	tc
upper	base	base	self/base-typ	90.6	90.6
upper	base	base	self/no-base	90.6	90.6
upper	base	base	neg/base-typ	90.6	90.6
upper	base	base	neg/no-base	90.6	90.6
upper	s2	ep0	self/base-typ	92.6	92.6
upper	s2	ep0	self/no-base	92.6	92.6
upper	s2	ep0	neg/base-typ	92.6	92.6
upper	s2	ep0	neg/no-base	92.6	92.6
upper	s2	ep1	self/base-typ	90.9	90.9
upper	s2	ep1	self/no-base	90.9	90.9
upper	s2	ep1	neg/base-typ	90.9	90.9
upper	s2	ep1	neg/no-base	90.9	90.9
upper	s2	ep2	self/base-typ	90.2	90.2
upper	s2	ep2	self/no-base	90.2	90.2
upper	s2	ep2	neg/base-typ	90.2	90.2
upper	s2	ep2	neg/no-base	90.2	90.2
upper	s4	ep0	self/base-typ	95.5	95.5
upper	s4	ep0	self/no-base	95.5	95.5
upper	s4	ep1	self/base-typ	95.5	95.5
upper	s4	ep1	self/no-base	95.5	95.5
upper	s4	ep2	self/base-typ	96.2	96.2
upper	s4	ep2	self/no-base	96.2	96.2
multi	base	base	self/base-typ	88.6	88.6
multi	base	base	self/no-base	88.6	88.6
multi	base	base	neg/base-typ	88.6	88.6
multi	base	base	neg/no-base	88.6	88.6
multi	s2	ep0	self/base-typ	90.4	90.4
multi	s2	ep0	self/no-base	90.4	90.4
multi	s2	ep0	neg/base-typ	90.4	90.4
multi	s2	ep0	neg/no-base	90.4	90.4
multi	s2	ep1	self/base-typ	92.0	92.0
multi	s2	ep1	self/no-base	92.0	92.0

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Data	Setting	Epoch	Eval	raw	tc
multi	s2	ep1	neg/base-typ	92.0	92.0
multi	s2	ep1	neg/no-base	92.0	92.0
multi	s2	ep2	self/base-typ	94.1	94.1
multi	s2	ep2	self/no-base	94.1	94.1
multi	s2	ep2	neg/base-typ	94.1	94.1
multi	s2	ep2	neg/no-base	94.1	94.1
multi	s4	ep0	self/base-typ	98.9	98.9
multi	s4	ep0	self/no-base	98.9	98.9
multi	s4	ep1	self/base-typ	99.5	99.5
multi	s4	ep1	self/no-base	99.5	99.5
multi	s4	ep2	self/base-typ	99.8	99.8
multi	s4	ep2	self/no-base	99.8	99.8

Table 3: HumanEval TRAIN-set val acc ($\times 100$) per-epoch dynamics, raw vs tc gen variant. gemma-4-31b, s2 (RankAlign) & s4 (self-tc), base/ep0/ep1/ep2, both datasets. Means over 82 problems, N=50 stratified candidates each.

Data	Setting	Epoch	Eval	raw	tc
upper	base	base	self/base-typ	70.0	70.0
upper	base	base	self/no-base	70.0	70.0
upper	base	base	neg/base-typ	70.0	70.0
upper	base	base	neg/no-base	70.0	70.0
upper	s2	ep0	self/base-typ	74.0	74.0
upper	s2	ep0	self/no-base	74.0	74.0
upper	s2	ep0	neg/base-typ	74.0	74.0
upper	s2	ep0	neg/no-base	74.0	74.0
upper	s2	ep1	self/base-typ	74.0	74.0
upper	s2	ep1	self/no-base	74.0	74.0
upper	s2	ep1	neg/base-typ	74.0	74.0
upper	s2	ep1	neg/no-base	74.0	74.0
upper	s2	ep2	self/base-typ	72.0	72.0
upper	s2	ep2	self/no-base	72.0	72.0
upper	s2	ep2	neg/base-typ	72.0	72.0
upper	s2	ep2	neg/no-base	72.0	72.0
upper	s4	ep0	self/base-typ	76.0	76.0
upper	s4	ep0	self/no-base	76.0	76.0
upper	s4	ep1	self/base-typ	80.0	80.0
upper	s4	ep1	self/no-base	80.0	80.0
upper	s4	ep2	self/base-typ	80.0	80.0
upper	s4	ep2	self/no-base	80.0	80.0
multi	base	base	self/base-typ	74.0	74.0
multi	base	base	self/no-base	74.0	74.0
multi	base	base	neg/base-typ	74.0	74.0
multi	base	base	neg/no-base	74.0	74.0
multi	s2	ep0	self/base-typ	76.0	76.0
multi	s2	ep0	self/no-base	76.0	76.0
multi	s2	ep0	neg/base-typ	76.0	76.0
multi	s2	ep0	neg/no-base	76.0	76.0
multi	s2	ep1	self/base-typ	74.0	74.0
multi	s2	ep1	self/no-base	74.0	74.0
multi	s2	ep1	neg/base-typ	74.0	74.0
multi	s2	ep1	neg/no-base	74.0	74.0
multi	s2	ep2	self/base-typ	78.0	78.0
multi	s2	ep2	self/no-base	78.0	78.0
multi	s2	ep2	neg/base-typ	78.0	78.0
multi	s2	ep2	neg/no-base	78.0	78.0

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Data	Setting	Epoch	Eval	raw	tc
multi	s4	ep0	self/base-typ	88.0	88.0
multi	s4	ep0	self/no-base	88.0	88.0
multi	s4	ep1	self/base-typ	80.0	80.0
multi	s4	ep1	self/no-base	80.0	80.0
multi	s4	ep2	self/base-typ	84.0	84.0
multi	s4	ep2	self/no-base	84.0	84.0

Table 4: HumanEval TRAIN-set Pearson ($\times 100$) per-epoch dynamics, raw vs tc gen variant. gemma-4-31b, s2 (RankAlign) & s4 (self-tc), base/ep0/ep1/ep2, both datasets. Means over 82 problems, N=50 stratified candidates each.

Data	Setting	Epoch	Eval	raw	tc
upper	base	base	self/base-typ	39.5	11.2
upper	base	base	self/no-base	39.5	11.2
upper	base	base	neg/base-typ	39.5	30.0
upper	base	base	neg/no-base	39.5	30.0
upper	s2	ep0	self/base-typ	51.1	22.7
upper	s2	ep0	self/no-base	51.1	27.2
upper	s2	ep0	neg/base-typ	51.1	30.7
upper	s2	ep0	neg/no-base	51.1	49.9
upper	s2	ep1	self/base-typ	50.0	20.8
upper	s2	ep1	self/no-base	50.0	26.2
upper	s2	ep1	neg/base-typ	50.0	32.3
upper	s2	ep1	neg/no-base	50.0	47.4
upper	s2	ep2	self/base-typ	51.1	18.6
upper	s2	ep2	self/no-base	51.1	27.4
upper	s2	ep2	neg/base-typ	51.1	25.5
upper	s2	ep2	neg/no-base	51.1	46.9
upper	s4	ep0	self/base-typ	53.3	40.3
upper	s4	ep0	self/no-base	53.3	36.3
upper	s4	ep1	self/base-typ	57.2	54.4
upper	s4	ep1	self/no-base	57.2	45.9
upper	s4	ep2	self/base-typ	57.3	57.8
upper	s4	ep2	self/no-base	57.3	52.5
multi	base	base	self/base-typ	15.8	19.4
multi	base	base	self/no-base	15.8	19.4
multi	base	base	neg/base-typ	15.8	19.4
multi	base	base	neg/no-base	15.8	19.4
multi	s2	ep0	self/base-typ	53.3	57.8
multi	s2	ep0	self/no-base	53.3	57.1
multi	s2	ep0	neg/base-typ	53.3	59.4
multi	s2	ep0	neg/no-base	53.3	47.0
multi	s2	ep1	self/base-typ	59.8	59.8
multi	s2	ep1	self/no-base	59.8	60.8
multi	s2	ep1	neg/base-typ	59.8	62.5
multi	s2	ep1	neg/no-base	59.8	52.9
multi	s2	ep2	self/base-typ	57.3	61.5
multi	s2	ep2	self/no-base	57.3	56.3
multi	s2	ep2	neg/base-typ	57.3	62.1
multi	s2	ep2	neg/no-base	57.3	43.9
multi	s4	ep0	self/base-typ	44.9	70.5
multi	s4	ep0	self/no-base	44.9	61.2
multi	s4	ep1	self/base-typ	47.4	72.1
multi	s4	ep1	self/no-base	47.4	60.4
multi	s4	ep2	self/base-typ	53.6	79.8
multi	s4	ep2	self/no-base	53.6	65.0

